

Original LCTRSs

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times int \Rightarrow State \\ \text{noncrit0} : Loc \\ \text{wait0} : Loc \\ \text{crit0} : Loc \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit0}, p1, b0, b1, x) \rightarrow \text{state}(\text{wait0}, p1, b01, b1, x1) \quad [b01 \wedge x1 = 1] \\ 2 : & \text{state}(\text{wait0}, p1, b0, b1, x) \rightarrow \text{state}(\text{crit0}, p1, b0, b1, x) \quad [x = 0 \vee \neg b1] \\ 3 : & \text{state}(\text{crit0}, p1, b0, b1, x) \rightarrow \text{state}(\text{noncrit0}, p1, b01, b1, x) \quad [\neg(b01)] \\ 4 : & \text{state}(p0, \text{noncrit1}, b0, b1, x) \rightarrow \text{state}(p0, \text{wait1}, b0, b11, x1) \quad [b11 \wedge x1 = 0] \\ 5 : & \text{state}(p0, \text{wait1}, b0, b1, x) \rightarrow \text{state}(p0, \text{crit1}, b0, b1, x) \quad [x = 1 \vee \neg b0] \\ 6 : & \text{state}(p0, \text{crit1}, b0, b1, x) \rightarrow \text{state}(p0, \text{noncrit1}, b0, b11, x) \quad [\neg(b11)] \\ 7 : & \text{state}(p0, p1, b0, b1, x) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit0}, \text{crit1}, b0, b1, x) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \ 9 : \ \text{state}(\text{noncrit0}, \text{noncrit1}, \text{false}, \text{false}, 0) \Rightarrow \text{success} \ }$$

BV-LCTRS Version

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times int \Rightarrow State \\ \text{noncrit0} : Loc \\ \text{wait0} : Loc \\ \text{crit0} : Loc \\ \text{noncrit1} : Loc \\ \text{wait1} : Loc \\ \text{crit1} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(\text{noncrit0}, p1, b0, b1, x) \rightarrow \text{state}(\text{wait0}, p1, b01, b1, x1) \quad [b01 \wedge x1 = 1] \\ 2 : & \text{state}(\text{wait0}, p1, b0, b1, x) \rightarrow \text{state}(\text{crit0}, p1, b0, b1, x) \quad [x = 0 \vee \neg b1] \\ 3 : & \text{state}(\text{crit0}, p1, b0, b1, x) \rightarrow \text{state}(\text{noncrit0}, p1, b01, b1, x) \quad [\neg(b01)] \\ 4 : & \text{state}(p0, \text{noncrit1}, b0, b1, x) \rightarrow \text{state}(p0, \text{wait1}, b0, b11, x1) \quad [b11 \wedge x1 = 0] \\ 5 : & \text{state}(p0, \text{wait1}, b0, b1, x) \rightarrow \text{state}(p0, \text{crit1}, b0, b1, x) \quad [x = 1 \vee \neg b0] \\ 6 : & \text{state}(p0, \text{crit1}, b0, b1, x) \rightarrow \text{state}(p0, \text{noncrit1}, b0, b11, x) \quad [\neg(b11)] \\ 7 : & \text{state}(p0, p1, b0, b1, x) \rightarrow \text{success} \\ 8 : & \text{state}(\text{crit0}, \text{crit1}, b0, b1, x) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \ 9 : \ \text{state}(\text{noncrit0}, \text{noncrit1}, \text{false}, \text{false}, 0) \Rightarrow \text{success} \ }$$